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Team boundary-spanning activities and performance of technology transfer organizations: evidence from China

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Abstract
This study performs an in-depth analysis of the causes and consequences of team boundary-spanning activities in the context of Chinese technology transfer, by addressing three related research questions: what are the drivers of teams' boundary-spanning activities? How do teams conduct boundary-spanning activities? What is the impact of teams' boundary-spanning activities? Based on the "input–moderator–output–input" model of team effectiveness theory, this study firstly explores which factors drive technology transfer teams to engage in boundary-spanning activities. Then, it examines the effects that such activities have on technology transfer performance, both directly and indirectly, through a set of mediating factors. Finally, it explores whether external environmental conditions play any moderating role in the relationship between team boundary-spanning activities and team performance. The empirical analysis is based on original primary survey data collected from a representative sample of organizations involved in the Chinese technology transfer system. The study both contributes the team effectiveness theory applied to the specific context of technology transfer and offers practical suggestions to managers of technology transfer intermediaries.

Keywords Technology transfer · China · Team effectiveness · Team boundary-spanning activities · Team performance

JEL Classification O30 · 032 · O39

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1 Introduction

Technology transfer institutions play a key role in enabling the transfer of scientific and technological knowledge between organizations (both companies and universities), limiting uncertainties and information asymmetries among parties (Agrawal 2001; Alexander and Martin 2013; Bercovitz et al. 2001; Comacchio et al. 2012). They play the important role of "bridging mechanisms" for the management of mixed areas between academia, industry, government, financial institutions and other independent actors (Howells 2006; Phan and Siegel 2006). Scholars have largely studied the dynamics of the technology transfer and have highlighted the most critical barriers that limit an effective flow of knowledge between university and industry. Among others, factors like cultural differences (Bjerregaard 2010), institutional settings (Bruneel et al. 2010), and geographic distance (D'Este et al. 2013) have been considered as the most relevant obstacles. The presence of such barriers has made the need to improve performance of technology transfer processes particularly urgent (Bruneel et al. 2010; Wu 2010).

Some of the solutions proposed include the development of technology transfer intermediaries with proper managerial skills and enhanced capabilities to manage boundary-spanning activities (Andersen and Kragh 2015; Mangematin et al. 2014). In fact, these organizations are responsible for building bridges between providers of scientific and technological knowledge (i.e., university researchers and departments) and external stakeholders (among which companies, individual entrepreneurs, and venture capitalists) (Hellström and Jacob 2003; Siegel et al. 2007), which operate in different contexts and pursue different institutional goals. Because of these differences, boundary-spanning activities often entail the translation of context-specific and complex knowledge and/or co-production of new knowledge (Howells 2006; Zhao and Anand 2013).

In the case of technology transfer intermediaries, the study of boundary-spanning activities has usually been addressed by taking an organizational perspective (Chau et al. 2017; Comacchio et al. 2012; Huyghe et al. 2014). Only rarely the analysis has been conducted at the individual or team level (e.g., Taheri and van Geenhuizen 2016). However, the role of matching the needs of suppliers and demanders of scientific and technological knowledge and of creating a linkage among them is often demanded to teams within technology transfer organizations. In turn, teams' social capital affects the amount and characteristics of social interactions established with external parties and eventually the outcomes of technology transfer processes (Joshi et al. 2009). It is therefore at the team level that boundary-spanning activities should be analyzed to address the problem of performance of technology transfer intermediaries. In fact, in the relationship between technology suppliers and demanders, technology transfer teams use external connections and boundary-spanning activities to achieve their goals (Marrone et al. 2007). Technology transfer teams provide a unique means of coordination and play a crucial role in establishing connections, as demonstrated by prior literature focusing on team-based organizational structures (Ancona and Caldwell 1992a, b; Marrone et al. 2007).

Although prior studies have shown that team boundary-spanning (TBS) activities are important to the success of organizations (Marrone 2010), at the best of our knowledge little attention has been paid by existing literature to a systematic examination of TBS activities in the specific context of technology transfer (e.g., Crupi 2020). Accordingly, this paper focuses on boundary-spanning activities promoted by technology transfer teams and investigates the internal and external factors that play a relevant role in this respect. The study addresses the following research questions: What are the drivers of teams' boundary-spanning activities? How do teams conduct boundary-spanning activities? What is the impact of teams' boundary-spanning activities?

A methodological exercise has been conducted focusing on China as a research setting. What makes China an interesting empirical framework is the vigorous development imposed by the government to technology transfer institutions, through the founding of government-established productivity promotion centers, university technology transfer offices, and industrial technology transfer centers. Overall, China has more than 3000 technology transfer institutions, 453 of which are national organizations. Thus, China represents a natural testbed for analyzing the behavior of technology transfer intermediaries and the impact of TBS activities. The empirical analysis has been conducted on original survey data collected from 402 technology transfer teams belonging to 247 Chinese technology transfer institutions (both government-funded institutions and independent technology transfer organizations).

In turn, this paper contributes to the stream of literature of team-based boundary-spanning activities by exploring the meso-level in the broader research field of organizational behavior. Specifically, consistent with Hackman (2003, p. 907), this study "(1) enrich(es) understanding of one's focal phenomena, (2) help(s) one discover non-obvious forces that drive those phenomena, (3) surface(s) unanticipated interactions that shape an outcome of special interest, and (4) inform(s) the choice of constructs in the development of actionable theory." Furthermore, the multilevel research model adopted by this study combines multiple theoretical perspectives to understand why teams engage in boundary-spanning activities, how they engage in specific boundary-spanning activities, and the consequences of specific boundary-spanning activities. Finally, the study extends the research field of TBS activities to the "IMOI" theoretical model of team effectiveness (Ilgen et al. 2005), which provides an important reference for exploring the endogenous mechanisms of the consequences of TBS activities.

2 Literature review

2.1 Boundaries and boundary-spanning activities

Previous studies deeply explored the idea of boundaries as interfaces between the internal organization system and the external environment with which the company cooperates (Aldrich and Herker 1977; Friedman and Podolny 1992). Hernes (2004) argues that boundary characteristics reflect the organization's nature, and organizational boundaries are the main coordinating mechanism of the organization. The role of boundaries relative to the external environment is twofold. On the one hand, in a cooperative system, the interface has the function of buffer and protection, separating the company from the environment. It gives the organization a certain degree of closeness and independence, thereby reducing the interference and influence of various external factors. On the other hand, in case of a conflictual situation in which the organization might have frictions with other external actors, boundaries protect undesired exchange of inputs and outputs and hinder the flow of resources and knowledge between the organization and the outside (Schotter et al. 2017).

Within organizational theory, boundary-spanning refers to the effort of a unit of work (such as a team) to connect to external resources (Ancona 1990; Marrone 2010). Externally oriented activities include managing changing customer needs, negotiating project scope, and obtaining critical information resources from outside units (Marrone et al. 2007). Such activities help companies develop links with relevant units of their external environment, manage interactions, pursue firm's objectives and improve performance. In terms of technology transfer, boundary-spanning activities facilitate knowledge flows through active networking, partnership and collaborative learning (Howells 2006; Marrone et al. 2007; Meyer and Kearnes 2013).

While previous studies addressed the concept of boundary-spanning activities mostly from the point of view of firms, Joshi et al. (2009) analyzed boundary-spanning as a phenomenon at the team level. It is expressed as a "shared team ownership", which originates from the experience, understanding, attitude, values, cognition, or behavior of individual team members. External-oriented activities include the diversity of team access to information, the opportunity to negotiate project expectations, the need to improve team coordination and performance. All together, these activities have a significant impact on organizational learning, innovation, and job performance. Thus, boundary-spanning activities at the team level include all "team's actions to establish linkages and manage interactions with parties in the external environment" (Marrone 2010, p. 914), and are intended to help teams and actors connected to them in reaching targets and performing tasks, especially within innovation-related projects (Belso-Martínez et al. 2020; Brunetta et al. 2020; Edmondson and Harvey 2018).

As for technology transfer, teams perform boundary-spanning activities between technology suppliers and demanders at different levels, from the single center to the entire regional innovation system (Cesaroni and Piccaluga 2016; Comacchio et al. 2012). TBS activities play a central role in improving team performance, allowing teams to acquire the resources and knowledge necessary to coordinate and complete tasks (Luciano et al. 2018). The external activities also offer indications of the technology transfer strategy pursued by the center (Corsino et al. 2018; Fitzgerald and Cunningham 2016). Through mission statements, technology transfer teams signal to institutional stakeholders, from academia, entrepreneurial environment and government (Siegel et al. 2003). As stressed by David and David (2003), mission statements are incisive goal statements that differentiate an organization from other similar organizations.

2.2 Theoretical models of team boundary-spanning activities

Although previous theoretical and empirical studies addressed teams' role in boundary-spanning activities, the literature has yet to fully explore the causes and consequences that push teams to engage in such activities (Brion et al. 2012; Choi 2002; Edmondson 1999; Knapp 2010; Marrone et al. 2007; Van den Bossche et al. 2006). For long, one of the most used theoretical models to address team effectiveness has been the input–process–output (I–P–O) model, according to which available inputs affect team's processes, which in turn affect team's outcomes. However, scholars pointed out some weaknesses of the model, such as Ilgen et al. (2005), who considered the team effectiveness framework of the I–P–O model flawed and inadequate for three main reasons. First, many of the mediating factors that pass the effects of input factors to the resulting elements cannot be called processes but rather are some emergent cognitive or affective states; second, the I–P–O model framework limits the research of team effectiveness to a simple single-loop linear path from input to output, ignoring the existence of feedback paths, which may limit the further development of team research; third, the I–P–O model framework considers that the main effects are a linear process of successive influences, with the input affecting the process and the process affecting the output in return, even though there may be an interaction between the two main effects of input and process, or between the process factors and the emerging state factors.

In turn, Ilgen et al. (2005) suggest the adoption of the alternative input–mediator–output–input (IMOI) model. In this model, input factors include team members (combination factors), the team, the organization and the environment (three levels). "M" instead of "P" represents the mediating factors, such as the team process and the sudden state of birth. The team process includes the transformation, execution and interpersonal processes; the inrush status includes the mediation mechanisms of recognition and emotion, such as team confidence, empowerment, atmosphere, cohesion, effectiveness, member trust, collective cognition, and so on. "O" represents team effectiveness. Finally, the last "I" represents the input factor of team output at a certain stage and the effectiveness of the next phase of the team, taking into account the time effect, adding feedback, and reflecting the dynamic changes from the output side to the input or process factors.

Overall, with respect to the I–P–O model, the IMOI model is better able to represent the causes and consequences of a team's actions. Accordingly, the following sections explore antecedents (inputs), mediators and outcomes (performance effects) of TBS activities. Figure 1 reports the theoretical framework adopted by the study.

2.3 Antecedents of team boundary-spanning activities

One of the research questions addressed in this study is why technology transfer teams need to carry out boundary-spanning activities. Scholars generally agree that task characteristics are the dynamic factors that drive the team to a boundary, and the dependence of the team on the task is determined by the position of the team in the internal organization workflow (Choi 2002; Joshi and Roh 2009). Task characteristics determine both team positioning during task execution and the need of information exchange, which, in turn, determines team dependence on external actors (Druskat and Wheeler 2003). When task complexity is high and the degree of dependence from external actors is strong, the tension between task time pressure and resource shortage pushes the team to spontaneously recur to boundary-spanning activities. In doing so, the TBS behavior influences performance.

As suggested by prior studies (Dutton et al. 2002; Joshi and Roh 2009; Maitlis 2005), activities are task-based, and the team chooses the external activity strategy according to the task characteristics and conditions. In particular, the most relevant task pre-factors that trigger effects on team TBS activities are task dependence, complexity and time pressure.

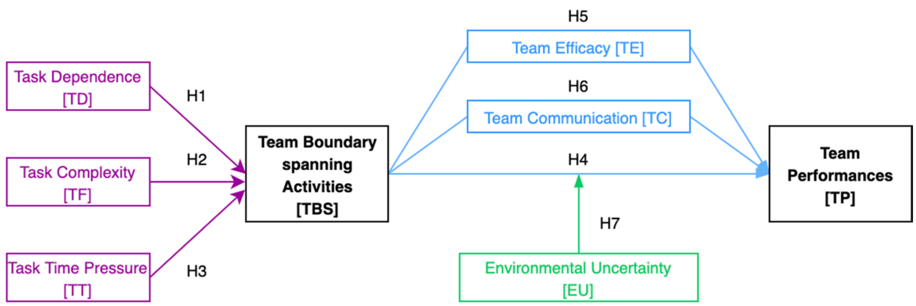

Fig. 1 Theoretical framework

2.3.1 Task dependence

Team external dependence concerns the fact that teams obtain resources (e.g., materials, services, knowledge, labor force and financial assets) and interchange outputs with the external environment (Demircioglu and Audretsch 2018; Wang et al. 2018). Previous literature is weak regarding how TBS behavior is affected by task dependence, intended as the extent to which teams exchange resources with other teams to fulfill their objectives (Choi 2002). In general terms, TBS activities vary depending on the relevance given by the organization to the resources and their matching with the different inter-team (or interdepartmental) units and their common goals (Tsai 2000). According to the resource dependence theory (Hillman et al. 2009), resource exchange drives inter-team interaction on the basis of organizational internal relationships. To some extent, these mechanisms highlight how task dependence drives teams to participate in external boundary-spanning activities to obtain resources and information (Tsai 2002).

Thus, task factors may trigger TBS activities and push teams to seek external links. To achieve high-quality interactive effects, teams are even more willing to enhance their interactions and share resources when boundary-spanning activities are highly dependent on external resources, especially when tasks are vaguely defined. Hence, team members maintain contacts with the external environment to obtain real-time assistance, which strengthens the impact of TBS activities on performance (Litchfield et al. 2018; Lowik et al. 2016). If the degree of task dependence directly affects external boundary-spanning activities, then in the case of strong dependence the technology transfer team will make every effort to quickly and effectively span boundaries. Accordingly:

Hypothesis 1 The greater the degree of task dependence, the higher the team's propensity to promote boundary-spanning activities.

2.3.2 Task complexity

Task complexity relates to the perception of difficulty perceived by team components in accomplishing the task and determines whether the team will engage in boundary-spanning activities (Ancona 1990; Hayter et al. 2018). Complex tasks are usually expressed as the need to obtain and deal with more information or to be more process- (or element-) involved. Thus, the completion of complex tasks requires comprehensive knowledge and high-level skills. In the case of technology transfer, a highly complex task may trigger teams to participate in external boundary-spanning activities. For instance, Faraj and Yan (2009) argue that as task-related information-processing requirements becomes more complex and demanding, teams need to access external resources and to promote a stronger coordination between internal and external activities. By engaging in TBS activities, the team is able to better understand the external environment and needs, and to access information and resources. Additionally, through boundary-spanning activities, teams can increase key stakeholders' understanding of project progress and performance, obtaining an objective evaluation of teams' performance.

In the case of complex tasks, teams face high uncertainty and environmental pressure; thus, they tend to search for similar knowledge from opponents to solve technical problems. Task complexity forces teams to gain expertise and information from the outside (Hargadon 1998), which includes technical knowledge related to projects and general information.

In tasks characterized by high complexity, the presence of unconventional problems and the numerous steps in the process of execution make the application of standardized procedures challenging. In turn, teams often need to carry out TBS activities to coordinate internal and external relations and obtain additional resources related to the task through border investigation (Bresman 2010). Accordingly:

Hypothesis 2 The greater the task complexity, the higher the team's propensity to promote boundary-spanning activities, and the greater the exchange of knowledge and other resources with external actors.

2.3.3 Task time pressure

Task time pressure refers to the extent to which personnel perceive they do not have adequate time to finish a task (Markus and Oldham 2006). With the approaching of a task's deadline, the team might looking for help from external leaders or reorganize and improve processes or systems to perform tasks efficiently (Chung and Jackson 2013). The position of the team in the network can affect the efficiency of resource flows; thus, the time urgency of completing a task can also affect TBS activities. When team members are aware that the deadline for a task is approaching, it is highly likely that they will seek the assistance of senior leaders to obtain critical resources and support. In addition, when the perceived time pressure is high, the team may negotiate with external customers to postpone tasks or achieve work goals by organizing workflow and systems and by dividing the labor and collaborating with external stakeholders. In the context of time pressure on team tasks, actors achieve common goals by coordinating external cooperative actions and standardizing the process (Oehler et al. 2018). In turn, it is possible to claim:

Hypothesis 3 The greater the task time pressure, the higher the propensity of teams to promote boundary-spanning activities.

2.4 Influence of team boundary-spanning activities on team performance

Prior literature shows that team boundary-spanning activities have a significant positive relationship with team performance (Horwitz and Horwitz 2007; Marrone et al. 2007; Tziner 1985). For instance, Ancona and Caldwell (1992a) identified three different types of external activities of new product development teams: ambassador activities, task coordinator activities, and scouting activities. Among them, ambassador activities significantly affect the initial budget and team performance; task coordinator activities positively influence team's long-term innovation performance; and the external communication frequency negatively predicts the team's initial budget and task scheduling performance. Other studies further confirm that the frequency of a team's external communication has a positive impact on both team performance and innovation efficiency, as evaluated by managers (Litchfield et al. 2018; Lowik et al. 2016).

Similarly, other empirical studies have confirmed the positive effect of TBS activities on performance. Workman (2005) found that the implementation of virtual teams is quickly increasing, particularly among transnational organizations that find global virtual teams a natural way to address their needs for global reach. Carson et al. (2007) showed that TBS behavior positively predicts the team survivability and team performance assessed by end users. Based on a survey on 31 students, Marrone et al. (2007) reported that TBS activities (including ambassador and task coordinator activities) aggregated at the team level predict client-rated team performance among teams of MBA students completing consulting projects. Finally, Faraj and Yan (2009) analyzed 64 software development teams and found that TBS activities promote team performance. Accordingly:

Hypothesis 4 The more the team's engagement in boundary-spanning activities, the higher the team performance.

2.5 Mediation effects

The adoption of the IMOI model framework to explore the causes and consequences of TBS activities in technology transfer allows to emphasize the critical role played by a set of factors that mediate the relationship between antecedents of TBS activities and team performance. Specifically, this study focuses on team efficacy and team communication.

2.5.1 Team efficacy

Team efficacy is intended as the team's reliance on its ability to effectively execute a given task (Gibson 1999; Gully et al. 2002; Kennedy et al. 2009), and represents a first important mediator between TBS activities and team outcome. In fact, through external activities teams can not only obtain information about the external environment, but also improve their ability to use (tacit and complex) knowledge and then improve team efficacy (Hansen 1999). Moreover, other studies, using a relationship model between team activity and team efficacy on the basis of structural contingency theory, show that TBS activities positively impact on team efficacy (Choi 2002; Edmondson 1999; Guzzo et al. 1993). Finally, others found that team efficacy and team potential are significantly and positively correlated with team performance (Gully et al. 2002; Stajkovic et al. 2009). Accordingly:

Hypothesis 5 The more a team is engaged in boundary-spanning activities, the higher the team efficacy, which in turn leads to greater team performance.

2.5.2 Team communication

Effective team communication is the core factor ensuring a team's operationality. As prior studies show Sivasubramaniam et al. (2012), often teams fail because of troubles in communication and coordination between members rather than because of a lack of competencies. Communication plays a mediating role between boundary-spanning activities and outcomes (Zhang et al. 2015). The more team members communicate with key people outside the team, the more effective the team is Ancona and Caldwell (1992a). Thus, the communication process can affect team performance if the team internal function is improved due to boundary spanning activities (Choi 2002; Edmondson 2003). From the empirical point of view, the study by Lievens and Moenaert (2000) on financial service teams confirms that team communication can significantly promote team innovation performance. Additionally, Hirst and Mann (2004) established a model of the impact of R&D project leadership on project performance and supported the intermediary role of team communication with empirical data. Accordingly:

Hypothesis 6 Team boundary-spanning activities influence team performance by improving team communication.

2.6 Moderation effect of environmental uncertainty

Concerning the factors that may influence TBS activities, only few empirical studies have focused on aspects that moderate the relationship between TBS activities and team performance. Among these, Faraj and Yan (2009) studied 64 software development teams and pointed out that the relationship between TBS activities and team performance is moderated by task uncertainty and resource scarcity. Furthermore, besides internal factors, the profound impact of the external environment on the team has been increasingly recognized in theory and practice (Ancona et al. 2002; Choi 2002). For instance, Gladstein (1984) studied the effectiveness of a working group (team effectiveness) as a function of the implemented boundary management processes and found that the relationship between the latter and the former is regulated by the environmental uncertainty. Over the years, studies adopting a similar approach have confirmed the importance of environmental characteristics on the relationship between boundary-spanning behavior and performance (Ancona 1990; Choi 2002; Gibson and Dibble 2013). The basic idea is that when environmental uncertainty increases, the difficulty to properly forecast future events, to deal with many and instable external actors, or to plan an optimal response to uncertain and risky events, requires teams to improve both internal and external activities. TBS activities become in this case particularly urgent and relevant to cope with an uncertain and risky environment, which eventually leads to a greater impact of TBS activities on performance. In turn, this study considers the effect that environmental uncertainty may have as a moderating factor:

Hypothesis 7 Environmental uncertainty positively moderates the relationship between team boundary-spanning activities and team performance.

3 Methodology

3.1 Research setting

To test hypotheses, the study focuses on technology transfer intermediaries of the manufacturing industry in China at the team level. Selected teams are mainly located in project and cooperation departments of the corresponding technology transfer intermediaries and have at least 4 members.

At present, in China there are more than 3000 technology transfer intermediaries and, among them, 453 intermediaries (officially referred to as the "National Technology Transfer Demonstration Organizations") were approved by the Chinese Ministry of Science and Technology. In terms of regional distribution, 275 demonstration organizations (corresponding to 60.70% of the total number of organizations) are located in the eastern region, 81 (17.88%) in the central region, and the remaining 97 (21.42%) in the western region. From the complete list of demonstration organizations, we randomly selected 300 technology transfer intermediaries. In order to respect the geographical proportion of demonstration organizations among the three regions, 180 technology transfer agencies were drawn from 11 provinces and cities in the East, 54 technology transfer agencies from 8 provinces and cities in the central region, and 66 technology transfer agencies from the 12 provinces and cities in the West.

According to the size of each technology transfer organization and the actual number of technology transfer teams, for each organization belonging to the sample we randomly selected 1–5 teams, and invited the team leader and three team members to complete the survey in order to ensure consistency and objectivity of answers provided. After excluding incomplete or invalid questionnaires, a total of 402 sets of valid questionnaires were collected, totaling 1608 copies (4 copies per organization). A set of independent sample t-tests were run to exclude any respondent and non-response biases.

Scales used in the questionnaire have been drawn from prior literature and further validated via preliminary interviews with a small sub-sample of technology transfer teams. This procedure allowed us to test the reliability and validity of adopted measurement scales. Based on validated questionnaires, the survey was conducted from January 2, 2018 to March 31, 2018. All scales used in the questionnaires are based on the Likert scale and considered continuous variables.

3.2 Independent variables

Previously, qualitative methods were often used to assess the nature and extent of team external processes in their team samples (Ancona 1990; Edmonson 1999). More recently, some researchers have begun to quantify the boundary-spanning activities of team members and then aggregate them at the team level for analysis (Ancona and Caldwell 1992a; Marrone et al. 2007). Ancona and Caldwell (1992a, b) explored the impact of boundary-spanning behavior of new product development teams on team performance, and developed a boundary-spanning cooperation scale with 24 items, such as "the team will bear pressure from outside the team, so as not to interfere with the work of team members".

In this study, items related to TBS activities are based on the scale developed by Ancona and Caldwell (1992a) and adapted to the specific context of China's technology transfer.

3.2.1 Antecedents of boundary-spanning activities

3.2.1.1 Task dependence Prior researchers measured task dependence in a single dimension. A more comprehensive approach has been adopted by Campion et al. (1993), who considered teams composed of two or more members who were used to share goals embedded in the organizational environment. According to this approach, task dependence refers to the extent to which (technology transfer) teams exchange resources with other teams to accomplish task objectives. In this work we adopted a scale consistent with the study of Campion et al. (1993) and adapted to the Chinese context.

3.2.1.2 Task complexity This is an important factor affecting TBS activities. Research on this topic addressed three main areas related to task complexity: information processing and decision-making, task and job design, and goal setting. Various measurement scales have been developed accordingly. Among these, the scale developed by Jehn (1995) considers the construct of complexity as dependent upon five aspects related to the task, from repeatability to decomposability: tasks contain many changes, the main task is to solve complex problems, it is difficult to routinize work, much information or many alternatives are required, and tasks include many different elements. This scale has been adapted to the Chinese context and employed in this study.

3.2.1.3 Task time pressure The study applies the scale proposed by Brown and Miller (2000), who suggested that task time pressure causes anxiety to team components; hence, cost-effective problem-solving methods have to be implemented by teams to deal with it. Once the team recognizes that the remaining time to fulfill the task before deadline is limited, help from outside leaders might be sought or the whole process to perform the task could be redesigned and reorganized. With the intensification in task time pressure, prior knowledge (experience) has an increasingly important impact on people's choices, encouraging the team to search for or adopt external mature experience. Furthermore, in the face of urgent tasks, technology transfer teams may also adopt other approaches. First, through communication with leaders or customers, they can delay the project delivery time for consultation. Second, by coordinating with external stakeholders, relevant process nodes may reduce the quality management standards of external coordination and collaboration processes. Third, by actively collecting information on competitors or customers, they can find technical breakthroughs to solve similar problems or invite outside industry experts to collaborate and offer guidance and make suggestions.

3.3 Dependent variable

Performance can be understood as the result of a series of job-related activities conducive to the improvement of the organization's core technical competence. Prior literature has analyzed different aspects related to team performance measurement, such as self-evaluations, superior's ratings, peer ratings, multiple ratings, cross ratings, subordinate ratings, committee ratings, and 360-degree evaluations. In this study we adopted the scale developed by Müller and Turner (2007), who identified 10 criteria associated to performance through interviews with professionals, and proposed a project-success-oriented performance measurement scale. This scale comprehensively evaluates the performance of technology transfer teams and reflects the overall content of team performance.

3.4 Mediator variables

3.4.1 Team efficacy

Existing studies do not converge on a unique method to measure team efficacy. As shown by Guzzo et al. (1993), one of the most important methods presented in the literature is the self-efficacy evaluation of individual team members, albeit this method does not take into account the social impact process and social control process. Alternatively, team efficacy has also been evaluated as a result of the overall team efficacy based on team members' joint discussion, even though such a method does not fit properly into the case of numerous research objects or samples that are too large.

Based on these considerations, the present study applies the scale developed by Guzzo et al. (1993) and adapted to the Chinese context of technology transfer. It comprehensively evaluates the efficacy of team technology transfer and systematically reflects the team efficacy general value.

3.4.2 Team communication

Hirst and Mann (2004) established a model to measure the impact of R&D leadership on project performance according to the mediating role of team communication. The literature shows that effective team communication promotes team innovation, enhances employees' organizational commitment and personal performance, and improves the overall project performance. By following the same approach, this study uses the 13 items listed by Hirst and Mann (2004) to measure team communication.

3.5 Moderator variable

Based on contingency theory, many scholars have analyzed the regulatory role of environmental uncertainty by proposing alternative measurement scales. For instance, Duncan (1972) divided environmental uncertainty into two dimensions: complexity and dynamics. Complexity is intended as the number of issues impacting on management activities in the business environment; dynamics is intended as the frequency and extent of changes in such issues. Dwyer and Welsh (1985) counted environmental characteristics and identified 27 main factors. Through empirical analysis, two common dimensions were extracted: heterogeneity, as the level of environmental diversification, and variability, as the level of environmental changes related to demand, competition and other factors. Child (1997) divided environmental uncertainty into the constitutive elements of complexity, diversity and illiquidity. Dess and Beard (1984) addressed richness, dynamics and complexity as three dimensions of environmental uncertainty. Richness is intended as the level of difficulty in acquiring resources in a business environment and the amount of elements conducive to the survival and growth of the firm. Dynamics is indented as the speed and extent of changes in the relevant factors affecting the activities of firms within the environment. Complexity concerns the amount of items to handle in an environment. Sharfman and Dean (1991) divided the environmental characteristics into three components: complexity, dynamicity and resource scarcity. Complexity is intended as the number and heterogeneity (or diversity) of environmental factors that firms must address. Dynamicity is indented as the frequency or unpredictability of environmental changes. Resource scarcity is the level of difficulty in obtaining environmental resources. Finally, Justin Tan and Litsschert (1994) divided the environment into three dimensions: dynamic, complex and hostile (threat).

Considered the characteristics of this study and the peculiarities of the Chinese technology transfer context, environmental uncertainty faced by technology transfer teams in the Chinese context has been measured by using the Justin Tan and Litsschert's (1994) research scale.

All the scales used for the research are reported in "Appendix 1".

4 Results

4.1 Sample descriptive statistics

Table 1 shows the distribution of sampled technology transfer intermediaries and technology transfer teams at the regional level, by revealing that the geographical distribution of the sample mimics the corresponding distribution of the national population. Indeed, the sample is composed of 247 technology transfer organizations in total. The 135 organizations located in the eastern part account for 54.66% of the whole sample, 41 in the central part account for 16.56%, and 71 in the West account for 28.78%. Similarly, the regional distribution of the 453 national technology transfer demonstration organizations shows that 275 organizations are located in the eastern region (60.71%), 81 in the Middle (17.88%), and 97 (21.41%) in the West. Table 2 shows the basic characteristics of the technology

Table 1 Geographical distribution

Area	Number of intermediaries	Percentage (%)	Number of teams	Percentage (%)
East	135	54.66	253	62.94
Middle	41	16.56	71	17.66
West	71	28.78	78	19.40
Total	247	100	402	100

transfer organizations included in the final sample. State-owned units (N = 196, accounting for 79.35% of the sample) represent the largest fraction. From these organizations, we addressed 308 teams, which account for 76.62% of the whole sample.

Tables 3 and 4 show, respectively, descriptive statistics and Pearson correlation coefficients of the variables included in the theoretical model. Task dependence and TBS activities are strongly correlated ($r = 0.656$, $P < 0.01$), which provides preliminary support for the verification of hypothesis H1. Task complexity and TBS activities are also positively correlated ($r = 0.712$, $P < 0.01$), providing preliminary support for hypothesis H2. Task time pressure is positively correlated with TBS activities ($r = 0.724$, $P < 0.01$), preliminarily supporting hypothesis H3. TBS activities are also positively correlated with team performance ($r = 0.213$, $P < 0.01$), which provides preliminary support for hypothesis H4.

4.2 Antecedents of boundary-spanning activities

The correlation analysis above shows that task dependence, complexity and time pressure are significantly correlated with TBS activities. To verify hypotheses H1, H2 and H3, TBS activities were taken as the dependent variable, while task dependence, task complexity and time pressure as independent variables. At the same time, six potential influencing factors—i.e., organization scale, organization ownership, organization location, team size, team setting time, and team stage—k were considered as control variables. Table 5 shows the results of the hierarchical linear regression analysis.

Model 1–1 shows that control variables (i.e., organization scale, ownership, location, team size, team setting time, and team stage) have no significant impact on TBS activities. Comparing model 1–2 with model 1–1, it emerges that the explanatory power of regression model improves significantly by adding the independent variable task dependence ($\Delta R^2 = 0.03$, $P < 0.01$). Moreover, the regression coefficients of the independent variables in model 1–2 show that task dependence positively impact on TBS activities ($\beta = 0.174$, $P < 0.01$). This indicates that the more task dependent the team is, the more likely it is to conduct TBS activities, therefore supporting H1. Model 1–3 shows that, with respect to model 1–1, the explanatory power of the regression model improves significantly by adding the independent variable task complexity to the control variables ($\Delta R^2 = 0.045$, $P < 0.01$). Moreover, the regression coefficients of the independent variables in model 1–3 show that task complexity positively impacts on team boundary-spanning activities ($\beta = 0.212$, $P < 0.01$). This indicates that the higher the task complexity that the team has to face, the more likely it is to conduct TBS activities. Confronting model 1–4 with model 1–1, it emerges that the explanatory power of the regression model is significantly improved by adding the independent variable task time pressure to the control variables ($\Delta R^2 = 0.03$, $P < 0.01$). Moreover, the regression

Table 2 Intermediary description

	Statistical content (N = 24)	Meas. code	N. of enterprises	Percentage (%)	Number of teams (N = 402)	Percentage (%)
Ownership	State owned enterprises	1	196	79.35	308	76.62
	Non-state-owned enterprises	0	51	20.65	94	23.38
Scale	Below 100 people	1	245	99.19	397	98.76
	101–300 people	0	2	0.81	5	1.24
	301–500 people		0	0	0	0
	501–1000 people		0	0	0	0
	More than 1000 people		0	0	0	0

Table 3 Variables' descriptive statistics

	Variable	Minimum	Maximum	Mean	S.D
1	Organization scale	1.000	2.000	1.012	0.110
2	Organization ownership	1.000	2.000	1.233	0.423
3	Organization location	1.000	3.000	2.017	0.609
4	Team size	1.000	5.000	1.985	0.733
5	Team setting time	2.000	5.000	3.853	0.556
6	Team stage	1.000	4.000	1.885	0.637
7	Task dependence	4.000	7.000	5.702	0.703
8	Task complexity	4.000	7.000	5.776	0.717
9	Task time pressure	4.000	7.000	5.719	0.709
10	Team boundary-spanning activities	4.000	8.000	6.045	0.744
11	Team performance	4.000	7.000	6.220	0.560
12	Team efficacy	5.000	7.000	6.059	0.661
13	Team communication	4.000	7.000	6.207	0.607
14	Environmental uncertainty	5.000	7.000	6.273	0.661

coefficients of the independent variables in model 1–4 show that task time pressure positively impacts on TBS activities ($\beta = 0.174$, $P < 0.01$). This indicates that the higher the task time pressure the team is confronted with, the more likely it is to conduct TBS activities. Therefore, hypothesis H3 proposed in the study is supported. Finally, model 1–5 shows that the explanatory power of the model significantly improves by adding to the control variables the three independent variables (task dependence, task complexity and task time pressure) ($\Delta R^2 = 0.045$, $P < 0.01$). Task dependence, task complexity and task time pressure have significant positive effects on TBS activities ($\beta = 0.205$, $P < 0.01$; $\beta = 0.226$, $P < 0.01$; $\beta = 0.003$, $P < 0.01$), which further supports H1, H2, and H3.

4.3 The effect of boundary-spanning activities on performance

Results of the correlation analysis among the variables show a significant correlation between TBS activities and team performance. To verify H4, team performance is taken as dependent variable, and TBS activities as independent variable, along with six potential influencing factors (control variables)—i.e., the scale of the organization, the nature of institutional ownership, the location of the organization, the size of the team, the time of team establishment and the stage of the team. Results of the hierarchical linear regression analysis are shown in Table 6, model 2–1.

Model 2–1 shows that organization scale, organization ownership, organization location, team size, team setting time, team stage have no significant impact on team performance. Comparing model 2–2 with model 2–1, it emerges that the explanatory power of regression model significantly improves by adding the independent variable (TBS activities) to the control variables ($\Delta R^2 = 0.047$, $P < 0.01$). Moreover, the regression coefficients in model 2–2 show that TBS activities positively influence team performance ($\beta = 0.217$, $P < 0.01$). This shows that the more the team is involved in

Table 4 Correlation coefficient matrix between variables

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	1													
2	0.203**	1												
3	−0.003	0.216**	1											
4	0.462**	0.453**	0.229	1										
5	0.232**	0.271**	0.074	0.668**	1									
6	0.373**	0.376**	0.147**	0.788**	0.859**	1								
7	−0.026*	−0.070*	−0.014	−0.021	−0.044	−0.052	1							
8	−0.030	−0.016	0.003	−0.044	−0.074	−0.079	0.846	1						
9	−0.038	−0.006	0.014	−0.065	−0.079	−0.086*	0.755**	0.819**	1					
10	−0.026	−0.007	−0.036	−0.038	−0.035	−0.050	0.656**	0.712**	0.724**	1				
11	0.054	0.016	0.011	0.087*	0.120*	0.080	0.170**	0.204**	0.164*	0.213**	1			
12	0.060	0.018	−0.037	−0.039	−0.041	−0.015	0.624**	0.628**	0.607**	0.619**	0.252**	1		
13	−0.024	0.005	−0.024	−0.019	−0.043	−0.042	0.532**	0.462**	0.513**	0.495**	0.110*	0.584**	1	
14	0.042	−0.045	0.000	0.025	0.010	−0.006	0.195**	0.183**	0.182**	0.282**	0.093*	0.177**	0.094*	1

$*P<0.1$; $**P<0.05$; $***P<0.01$

Table 5 Effect of antecedents on independent variable

	Model 1–1	Model 1–2	Model 1–3	Model 1–4	Model 1–5
Organization scale	0.072	0.079	0.077	0.075	0.077
Organization ownership	–0.008	–0.009	–0.009	–0.011	–0.009
Organization location	0.003	0.004	0.003	0.002	0.002
Team size	0.014	0.012	0.012	0.015	0.012
Team setting time	0.073	0.074	0.075	0.075	0.075
Team stage	–0.046	–0.042	–0.040	–0.042	–0.040
Task dependence		0.08***			–0.009***
Task complexity			0.096***		0.102***
Task time pressure				0.078***	0.001***
R^2	0.019	0.049	0.064	0.049	0.064
Adjusted R^2	0.004	0.032	0.047	0.032	0.042
ΔR^2		0.030	0.045	0.030	0.045
F	1.304***	2.933**	3.857***	2.918***	2.989**

*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$

boundary-spanning activities, the higher the likelihood of greater team performance. Therefore, hypothesis H4 is supported.

4.4 The role of mediating variables

To investigate the mediating role of team efficacy in the relationship between TBS activities and team performance (H5), the procedure proposed by Baron and Kenny (1986) by means of hierarchical regression analysis has been implemented. Specifically, firstly, the impact of TBS activities on team performance has been checked. Results of this analysis have already been presented in the previous section (Table 6, model 2–2). Then, the impact of TBS activities on team efficacy has been checked as well (model 3–1). Finally, as a third step, it has been checked whether the impact of TBS activities on team performance is still significant or declining when TBS activities and team efficacy are simultaneously included in the regression model (Table 6, model 3–2).

Model 3–1 shows that TBS activities positively affect team efficacy and model 3–2 confirms that the explanatory power of the model is improved significantly ($\Delta R^2 = 0.026$, $P < 0.01$) by adding the mediating variable team efficacy. TBS activities still have a positive effect on team performance, but the path coefficient decreases from 0.145 to 0.060. This shows that TBS activities not only have a direct impact on team performance but also have an indirect effect on team performance through team efficacy. That is, team efficacy plays a mediating role in the relationship between TBS activities and team performance. Thus, H5 is assumed to be supported.

To verify the mediating role of team communication (H6), we again used the method proposed by Baron and Kenny (1986). Specifically, the first step has been to detect the effect of TBS activities on team performance (already tested, Table 6, model 2–2), the second has been to detect the effect of the same variable on team communication (Table 6, model 4–1), and the third step has been to check whether the impact of TBS activities on team performance is still significant or declining when TBS activities and team communication are simultaneously included in the regression model (Table 6, model 4–2).

Table 6 Effects of independent and mediating variables on dependent variable

	Model 2–1 Dep. Var: Team performance	Model 2–2 Dep. Var: Team performance	Model 3–1 Dep. Var: Team efficacy	Model 3–2 Dep. Var: Team performance	Model 4–1 Dep. Var.: Team communication	Model 4–2 Dep. Var.: Team performance
Organization scale	0.072	0.077	0.258	0.043	−0.042	0.077
Organization ownership	−0.008	−0.011	0.022	−0.014	0.006	−0.011
Organization location	0.003	0.005	−0.004	0.006	−0.006	0.005
Team size	0.014	0.014	−0.052	0.021	0.013	0.014
Team setting time	0.073	0.072	−0.051	0.079	−0.020	0.072
Team stage	−0.046	−0.041	0.071	−0.051	0.000	−0.041
TBS activities		0.145***	0.640***	0.060	0.374***	1.41***
Team efficacy				0.134***		
Team communication						0.010
R^2	0.019	0.066	0.396	0.092	0.247	0.066
Adjusted R^2	0.004	0.049	0.385	0.073	0.233	0.047
ΔR^2		0.047		0.026		0
F	1.304***	3.985***	36.879***	4.958***	18.377***	3.483***

$^*P < 0.1$; $^{**}P < 0.05$; $^{***}P < 0.01$

Results from model 4–2 show that TBS activities significantly and positively affect team communication, and model 4–3 reveals both that there is no significant change in the explanatory power of the model ($\Delta R^2 = 0.000$, $P > 0.1$), and that the coefficient associated to team communication is no more significant when this mediator variable is added to the model along with the independent variable (model 4–2). Therefore, H6 is not supported.

4.4.1 Moderation effect of environmental uncertainty

We use the hierarchical linear regression equation method to verify the moderating effect of environmental uncertainty on the relationship between TBS activities and team performance, controlling for potential variables that may affect team performance, such as organization scale, organization ownership, organization location, team size, team setting time, and team stage. Table 7 shows the results of this analysis. In calculating the interaction term, the independent variable and the moderating variable are centered around the mean in order to minimize possible collinearity between the interaction term and both the independent variable and the moderating variable. Maximum variance inflation factor (VIF) values of each model are all below the threshold of 10, hence indicating the absence of serious multicollinearity problems.

Results from model 5–4 show that the explanatory power of the model is improved significantly ($\Delta R^2 = 0.030$, $P < 0.01$) when the interaction term between the moderating variable and the main independent variable is introduced into the model. However, the corresponding regression coefficient is not significant. Therefore, environmental uncertainty does not play any significant moderating role in the relationship between TBS activities and team performance. That is, regardless of the degree of environmental uncertainty, the impact of TBS activities on team performance is not significantly strengthened or weakened. In turn, we deduce that H7 cannot be supported.

Table 7 The moderating effect of environmental uncertainty on the relationship between team boundary-spanning activities and team performance

	Model 5–1	Model 5–2	Model 5–3	Model 5–4
Organization scale	0.072	0.077	0.075	0.081
Organization ownership	−0.008	−0.011	−0.010	−0.011
Organization location	0.003	0.005	0.005	0.007
Team size	0.014	0.014	0.013	0.008
Team setting time	0.073	0.072	0.071	0.062*
Team stage	−0.046	−0.041	−0.041	−0.033
TBS activities		0.145***	0.140***	2.096***
Environmental uncertainty			0.019	1.908***
TBS activities * Environmental uncertainty				−0.253
F	1.304***	3.985***	3.517***	4.649***
R^2	0.019	0.066	0.066	0.096
Adjusted R^2	0.004	0.049	0.047	0.075
ΔR^2			0.000	0.030
VIF maximum	1.019	1.071	1.071	1.106

*$P < 0.1$; **$P < 0.05$; ***$P < 0.01$

Table 8 Summary of hypotheses' testing

	Hypothesis	Result
Hypothesis 1	Task dependency positively influences team boundary-spanning activities	Supported
Hypothesis 2	Task complexity positively influences team boundary-spanning activities	Supported
Hypothesis 3	Task time pressure positively influences team boundary-spanning activities	Supported
Hypothesis 4	TBS activities positively influences team performance	Supported
Hypothesis 5	TBS activities influences team performance by improving team efficacy	Supported
Hypothesis 6	TBS activities influences team performance by improving team communication	Non-supported
Hypothesis 7	Environmental uncertainty positively moderates the relationship between TBS activities and team performance	Non-supported

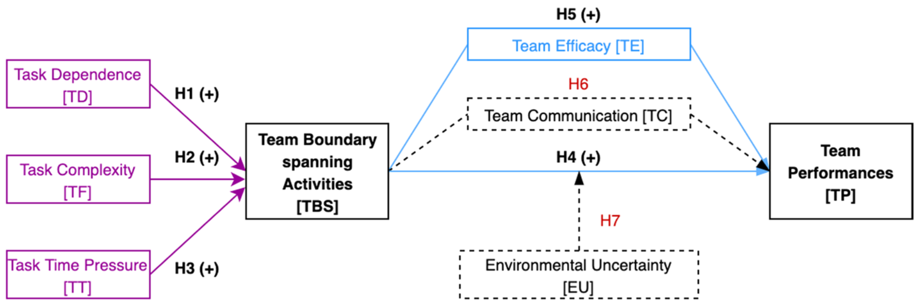

Fig. 2 Resulted framework

Table 8 lists the results of the various regression analyses, while Fig. 2 graphically shows the relationships between the variables after the analysis.

5 Discussion and conclusions

The study proposes a research model to investigate the role of TBS activities on team performance in the context of technology transfer in China and aims at contributing to the resource dependence theory by extending it to TBS activities. The empirical analysis conducted on a sample of teams belonging to Chinese technology transfer intermediaries confirms most of the proposed hypotheses. Specifically, regarding antecedents, obtained results support H1, H2, and H3, thus demonstrating that task dependence, complexity, and time pressure, respectively, have a positive effect on TBS activities. Furthermore, results also support H4, thus verifying that TBS activities have a positive effect on team performance in the context of Chinese technology transfer. In turn, this work confirms prior studies on the impact of TBS activities on team performance—which empirically demonstrated the existence of a positive relationship between the two variables (Ancona and Caldwell 1992b; Marrone et al. 2007)—and extends those results to the specific context of technology transfer—which had not been broadly investigated by prior literature. In this, the study also confirms that the "IMOI" model (Ilgen et al. 2005) related to the team effectiveness theory is also applicable to the context of technology transfer in China.

Concerning the mediating effect of team efficacy, obtained results support H5. It emerges that team efficacy plays a mediating role between TBS activities and performance in the context of technology transfer. By contrast, the mediating role of team communication is not verified by the study (H6 is not supported). Therefore, by taking both results into consideration, it emerges that the "IMOI" model of team effectiveness theory can only partly explain the effects of team boundary-spanning activities on team performance in the context of technology transfer.

Finally, as for the moderation effect of environmental uncertainty, obtained results do not support H7. Environmental uncertainty does not seem to affect positively the impact of TBS activities on team performance as expected. This outcome may be due to different motivations. First, the adopted measurement scale (adapted from Justin Tan and Litsschert 1994) may not fit perfectly the characteristics of the technology transfer context faced by sampled organizations, and may not capture perfectly some of the conditions that affect the relationship between TBS activities and team performance. In this case, alternative scales of environmental uncertainty may be used to test the sensitivity of obtained results to measurement issues. Second, since the study has been conducted on organizations belonging to the same country (China), which is strongly influenced by the prominent role of government, it could be that all sampled teams face similar environmental conditions with limited variability. As a matter of facts, all respondents to our survey graded environmental uncertainty between 5.0 and 7.0 (Table 3). To overcome this limitation, future studies may consider multi-country samples with organizations confronted with different external environments. Finally, obtained results may simply convey the message that environmental factors may not be applicable to the moderating effect of TBS activities on team performance in the context of technology transfer. Further investigations on different contexts and samples might be conducted to confirm or disconfirm our research outcomes.

5.1 Theoretical contribution

This study performed a deep analysis of the causes and consequences of TBS activities in the context of technology transfer and expanded the applicability of team effectiveness theory (Joshi and Roh 2009) to that context by adopting the promising perspective of the "IMOI" model (Ilgen et al. 2005). Furthermore, since TBS activities are situationally dependent and vary from country to country and for different industries, companies and teams, the analysis of specific contexts in which TBS activities may have a positive effect on team performance is of particular interest to management research. This represents a first contribution offered by the study as it expands our understanding about under which conditions the team effectiveness theory may be successfully applied in technology transfer.

Furthermore, although a few prior studies have already focused on boundary-spanning activities at the team level (Chau et al. 2017; Comacchio et al. 2012; Marrone et al. 2007; Taheri and van Geenhuizen 2016), the real motivations behind the latter have not been fully explored. In particular, less attention has been paid by prior literature to the analysis of how the characteristics of technology transfer tasks may affect TBS activities. Moreover, the empirical results of previous studies on the impact of TBS activities on team performance offer a puzzling panorama. What is acknowledged by prior literature, however, is that external boundary-spanning perspective provide guidance for teams to perform complex and urgent tasks with the help of their relationships with the external environment (Edmondson 1999). A complete picture that considers TBS activities from drivers to ultimate consequences is missing. Accordingly, by adopting the perspective of resource dependence theory, this study analyzed the impact of pre-task mechanisms of TBS activities (namely, task dependence, task complexity and task time pressure) on TBS. The findings answer another question raised at the beginning of this study—Why should teams carry out boundary-spanning activities?—and further expand research on antecedents of team boundary spanning and task characteristics (e.g., Choi 2002; Joshi and Roh 2009).

Moreover, by adopting the "IMOI" model of team effectiveness theory, the study shows the positive effect of TBS activities on team performance. The study also highlights the mediating role of team efficacy in this relationship, thus offering to scholars a relevant stimulus to promote further efforts to dig into team boundary-spanning activity interactions.

5.2 Managerial and policy implications

Results of this study can also be interpreted in terms of managerial and policy implications. In fact, in the global context of rapid development of technology transfer, this research provides guidance for the full implementation of technology transfer norms and the enhancement of technology transfer capabilities. It also provides a set of inspirations for technology transfer intermediary managers. First, technology transfer intermediaries should recognize the great importance of TBS activities because of the positive role played by team efficacy in promoting team performance. The mediating effect shown by our empirical findings is also applicable to different organization scales, ownership modes, locations, and team settings. Second, technology transfer intermediaries should strengthen top-down task management processes. Task dependence, task complexity and task time pressure are the driving factors enabling technology transfer intermediaries to engage in TBS activities. When highly dependent tasks are complex and time intensive, technology transfer intermediaries spontaneously carry out TBS activities. Technology transfer managers can judge whether such activities need to be carried out based on the recognition of task dependence, task complexity and task time. Third, technology transfer intermediaries should also enhance team efficacy of team members by implementing a flat organizational management structure. Moreover, they can form a management team composed of technology transfer professionals to improve members' boundary-spanning communication functions and facilitate communication with partners.

Finally, the findings from our study have important policy implications, especially regarding the complexity of the setting in which it is grounded. Indeed, in the last decades, the world is witnessing a massive improvement of Chinese technological and innovation capabilities. On the one hand, this improvement is fed by a monumental effort in terms of internal public R&D expenditures to upgrade Chinese companies' capabilities. On the other hand, it is fed by a systematic activity undertaken by Chinese firms and consisting in massive foreign technology acquisition and absorption, thanks to the Chinese government's policy to promote actions to increase technological capabilities of Chinese firms (Prud'homme et al. 2018). The importance of TBS activities for team performance should suggest to policymakers to promote policies to provide platforms and programs of collaboration through which Chinese technology transfer teams can cooperate with foreign partners to reduce the technological gaps or create a collaborative network. However, technology transfer only works in case of mutual benefits, so these policies should be incorporated within a more general technological collaboration framework to guarantee such a reciprocity.

5.3 Limitations and future research

This article is an exploratory study of TBS activities of technology transfer and the related mechanisms of action. Although some meaningful research results have been obtained, a few limitations inevitably exist due to the subjective ability and the objective resource constraints. Firstly, this study did not split TBS activities into their main components. This is a direction for future research, which might consider TBS activities not as a unitary group but as a diversified bundle of activities, each of which with its own specificities. To date, most of the literature has focused on behavioral studies in the coordination dimension, with few attempts to distinguish between the causes or consequences of different dimensions (Joshi and Roh 2009).

A second limitation is the use of cross-sectional data to test hypotheses. In fact, the use of longitudinal data would have allowed an examination of the changes in TBS activity over time and their relationships with team performance.

Thirdly, although the empirical analysis conducted in this study meets the basic requirements of sample size, obtained empirical results should be generalized with caution to contexts different from what analyzed. In fact, certain heterogeneity exists both among technology transfer organizations and among technology transfer teams within the same organization. Under these conditions, it may be difficult to use a general theoretical model to include the generality and characteristics of technology transfer teams of different team sizes and team phases. Future research may adopt more fine-tuned sampling methods in order to enhance data representativeness and reliability of research conclusions.

Finally, regarding the selection of moderating variables between TBS activities and team performance in technology transfer, we limited our analysis to the impact of environmental uncertainty. Of course, other moderating variables may affect the relationship between team activities and team performance, such as external dependence, internal conflicts in the organization and team resources. In turn, future research may explore these other dimensions to provide a more complete representation of the effects of TBS activities on team performance in the context of technology transfer.

Appendix 1

Team boundary-spanning activity measurement scale—Source: Adapted from Ancona and Caldwell (1992a)

TB1	Absorb outside pressures for the team so it can work free of interference
TB2	Protect the team from outside interference
TB3	Teams often reject too many requests from outsiders
TB4	Persuade other individuals that the team's activities are important
TB5	Scan the environment inside your organization for threats to the product team
TB6	Team show its image to the outside world
TB7	Persuade others to support the team's decisions
TB8	Acquire resources (e, g., money, new members, equipment) for the team

TB9	Report the progress of the team to a higher organizational level
TB10	Find out whether others in the company support or oppose your team's activities
TB11	Find out information on your company's strategy or political situation that may affect the project
TB12	Keep other groups in the company informed of your team's activities
TB13	Resolve design problems with external groups
TB14	Coordinate activities with external groups
TB15	Procure things which the team needs from other groups or individuals in the company
TB16	Negotiate with others for delivery deadlines
TB17	Review product design with outsiders
TB18	Find out what competing firms or groups are doing on similar projects
TB19	Scan the environment inside or outside the organization for marketing ideas/expertise
TB20	Collect technical information/ideas from individuals outside of the team
TB21	Scan the environment inside or outside the organization for technical ideas/expertise
TB22	Keep news about the team secret from others in the company until the appropriate time
TB23	Avoid releasing information to others in the company to protect the team's image or product it is working on
TB24	Control the release of information from the team in an effort to present the profile we want to show

Task dependence research scale—Source: Adapted from Campion et al. (1993)

TD1	I cannot accomplish my task without information or materials from other members of my team
TD2	Other members of my team depend on me for information or materials needed to perform their tasks
TD3	Without my team, jobs performed by team members are related to one another

Task complexity research scale—Source: Adapted from Jehn (1995)

TF1	The task contains many uncertain factories
TF2	The main job is to solve complex problems
TF3	It is difficult to routinize the work
TF4	It requires a lot of information or alternatives
TF5	It includes many different elements

Task time pressure measurement scale—Source: Adapted from Brown and Miller (2000)

TT1	The time limit for technology transfer projects is urgent
TT2	Members of the technology transfer team face a lot of work
TT3	Members of the technology transfer team have no time to do other things
TT4	Members of the technology transfer team always feel that time is too little

Team efficacy measurement scale—Source: Adapted from Guzzo et al. (1993)

TE1	This team has confidence in itself
TE2	This team believes it can become unusually good at producing high-quality work
TE3	This team expects to be known as a high-performing team
TE4	This team feels it can solve any problem it encounters
TE5	This team believes it can be very productive
TE6	This team can get a lot done when it works hard

TE7	The team's work efficiency is high
TE8	This team expects to have a lot of influence around here

Team communication measurement scale—Source: Adapted from Hirst and Mann (2004)

TC1	Team members have a clear understanding of project objectives
TC2	Project objectives are understood by all members
TC3	There is a lack of clarity concerning project priorities
TC4	Project objectives are clearly communicated to all members
TC5	The team receives clear feedback regarding the project's performance
TC6	Team members receive clear feedback regarding the quality of project work
TC7	Project information is shared across the team and is accessible to all
TC8	It is often difficult to get answers to important questions about my work
TC9	Team members have access to all the information required to do their work effectively
TC10	Team members have a clear understanding of the expectations of customer/funding agencies
TC11	The team discusses project objectives with customer/funding agencies
TC12	Customer/funding agencies provide clear directions concerning desired project outcomes
TC13	The team receives frequent feedback from customer/funding agencies

Environmental uncertainty measurement scale—Source: Adapted from Justin Tan and Litsschert (1994)

EU1	Environmental changes in our local market are intense
EU2	Our clients regularly ask for new products and services
EU3	In our local market, changes are taking place continuously
EU4	In a year, nothing has changed in our market
EU5	The demand of technology demanders is largely influenced by non-market factors such as social culture, political factors, social events and policy orientation
EU6	Technological standards of technology suppliers are largely influenced by factors such as social culture and government policies
EU7	Our organizational unit has relatively strong competitors
EU8	Competition in our local market is extremely high
EU9	Price competition is a hallmark of our local market
EU10	Clients' requirements are getting higher and higher
EU11	The competition between teams is becoming more and more intense
EU12	Competitors' behaviors become more and more various
EU13	It is more and more difficult to obtain resources

Performance of technology transfer team—Source: Adapted from Müller and Turner (2007)

TP1	End-user satisfaction with the project's product or service
TP2	Suppliers' satisfaction
TP3	Project team's satisfaction
TP4	Other stakeholders' satisfaction
TP5	Technology transfer fails to achieve its overall performance (function, budget, etc.)
TP6	Meeting user requirements
TP7	Transferred technology failed to achieve its intended technical performance
TP8	Reoccurring business with the client
TP9	Meeting the respondent's self-defined success factor
TP10	Meeting the project's purpose

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